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Dentistry and the Behavioral/Social Sciences: An Historical Overview

Lois K. Cohen[1]

The development and growth of the social and behavioral sciences in the field of dentistry perhaps can best be described through a brief history of the Federal Government's involvement in this area, since prior to the mid-1950s, before Federal support began, there was very little scientific behavioral literature in the dental field.

During the earliest stages of development in the 1950s, a relatively broad range of substantive issues was being examined, suggesting a wide variety of behavioral and social science disciplines potentially applicable to dentistry. These issues include research on psychophysiological aspects of oral functions, psychologic aspects of oral disease or malfunction conditions, individual behavior characteristics, therapeutic regimens, and public behaviors in health-care programs.

In an annotated bibliography on psychology in dentistry published in 1962, the literature fell into three categories (Borland *et al.,* 1962). The first category included anecdotal accounts of patients' experiences, followed by a set of recipes for influencing their behaviors. A small second group of references, although more consistent with accepted concepts in psychology, psychiatry, and similar disciplines, also reflected inadequate preparation in the behavioral sciences. An extremely small third group of articles, authored by sophisticated behavioral scientists, presented theoretical material that followed the rigorous demands of research methods.

Although a few early articles dealt with psychosomatic processes involved in some dental diseases and with the sociopsychological aspects

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of motivation to seek care, the majority concentrated on practical dentist—patient problems, notably fear and anxiety associated with both dental pain and treatment (Borland *et al.,* 1962). Public health research which focused on the impact of professional intervention on the oral health of patients rarely was found during the late 1950s and early 1960s. Most of the work was supported by individual investigators, primarily those with a psychological orientation, or by their respective institutions.

The Division of Dental Resources of the U.S. Public Health Service (USPHS) conducted a series of intramural studies in 1950 regarding the supply and distribution of dentists and auxiliaries, the mechanisms to enhance dentists' productivity, and the facilities and costs of dental-school education. In 1954, additional research studied the expenses for students of dentistry and dental hygiene, as well as the various types of prepayment mechanisms for dental care. The USPHS Division of Dental Public Health simultaneously was carrying out research on community water fluoridation and investigating public attitudes and opinions about dental care. In 1960 when these two divisions merged to become the Division of Dental Public Health and Resources, the initial social science research expanded to new areas that included research on educational training methods for dental personnel, evaluating community dental programs, and educating the public about dental health.

During Federal years 1965 and 1966, this division funded fellowship awards and made grants available for projects of community dental research among university scientists, resulting in the promotion and expansion of the "sociodental field" (a shorthand term for social and behavioral research in dentistry). During the initial years after the divisions merged, the social and behavioral science staff was strengthened by including sociologists, psychologists, sociopsychologists, an economist, and a cultural anthropologist. Their intramural multidisciplinary skills provided technical assistance and consultation to the extramural staff responsible for stimulating research grants and training programs.

Much of that early research focused on fluoridation, questioning why it was or was not instituted in communities. Although many answers were generated from this research, the division did not undertake systematic research on intervention strategies, partly because the investigators were chiefly interested in testing explanatory theories and then shifting these assumptions to other case histories, such as school bond issues. A cadre of scientists committed to the continuous study of oral health was difficult to build and slow to emerge. Another reason for failing to study intervention methods was fear that Federal involvement at the state or local level would violate states' rights. In 1961, despite the failure to move from descriptive and analytic research to intervention research, a

number of insights regarding fluoridation were reported (Paul *et al.*, 1961).

A second bibliography, published in 1967, revealed an increased reliance on research data pertaining to dental manpower recruitment, education, training, and fluoridation, allowing a decreased dependence on anecdotal case histories and "armchair' theorizing (Evans, 1967).

Today the problems of motivating a community to accept fluoridation and other preventive agents still remain, accompanied by a scarcity of current research suitable for helping community organizers to institute local action through administrative processes or political machinery.

As schools of dentistry began to employ educational psychologists as consultants on methods of dental education, curriculum development, and evaluation, the training of dental students to use auxiliaries, productive and permanent relations emerged between psychologists and members of the dental profession. As early as 1946, numerous failures among dental students caused an American Dental Association (ADA) committee to try to develop a screening device for incorporation in the admissions process. Their request for assistance from the University of Chicago started a long association between dentistry and experts in educational measurement. The subsequent development of the Chalk Carving Test and the Dental Aptitude Testing Program formed a significant link between behavioral science and dentistry (Klieger, 1976; Peterson, 1974).

When departments of community or preventive dentistry emerged in the United States in the 1960s, a focus on the epidemiology, prevention, and control of oral disease began to appear. This led to the analysis and description of patients' motivations to seek and accept preventive treatment. Studies of the characteristic practices among the nation's dentists were rarer than were studies of the public's oral health behavior, yet a few were completed. Incidentally, practically all of the research prior to the sixties was conducted by social or behavioral scientists without dental qualifications.

In the 1960s the Government became aware of the need for trained individuals who combined an understanding of dental problems and knowledge of the methods of social—behavioral research. Dentists were encouraged to pursue a degree in one of the social or behavioral sciences, and conversely social and behavioral scientists were supported for graduate studies of oral health problems. Both pre- and postdoctoral Federal monies helped individuals to develop these multidisciplinary skills.

The National Institute of Dental Research (NIDR) supported several predoctoral research training programs. The University of Pittsburgh's program focused on physiological psychology, and as their trainees moved on to other universities they generated conditions suitable for funding

grants in related areas. NIDR also supported a training program designed to produce Ph.D.'s in social psychology with a particular interest in dental problems.

The Division of Dental Health, formerly called the Division of Dental Public Health and Resources, also supported training and grants in the sociodental field until the early 1970s. One predoctoral research training program educated dentists in the social sciences and educated social scientists in dentistry, while another grant trained health educators for research in the oral health field or dental personnel for research in health education. More recently, the NIDR has supported a postdoctoral grant that focuses on health motivation and human behavior. While fewer resources are available currently for institutional support, individual postdoctoral fellowships exist for facilitating such training.

An organization called Behavioral Scientists in Dental Research (BSDR) was formed officially in 1968. This worldwide organization has grown to approximately 200 members who represent a variety of disciplines.

In 1972 the BSDR formally became an affiliate of the Federation Dentaire Internationale (FDI). A brief history of this development illustrates the internationalization of sociodental research. FDI had supported a session on social science at their 1966 meeting in Tel Aviv. In 1969, their Commission on Dental Practice adopted a resolution to sponsor a critical bibliography on social sciences and dentistry (Richards and Cohen, 1971), and when this publication was issued in 1971, approximately 60 behavioral and social scientists joined with dental colleagues to participate in an open session on "Social Sciences and Dentistry." Since that time, BSDR members have become consultants to this federation of national dental associations involved in evaluating scientific programs; in formulating scientific sessions containing behavioral and social science material; and in working with dental colleagues to develop a consensus about practical issues, such as dental health education or social aspects of dental appearance. This affiliation, perhaps more than any other, permitted international collaborative research to fluorish and enabled the results of the research to be disseminated and utilized—a rare phenomenon in any scientific field.

Sociodental research has burgeoned in the 1970s despite the termination of its support by the Division of Dentistry, previously known as the Division of Dental Health. Credible publications on this subject continue to include textbooks for students of the dental professions (Ayer and Hirschman, 1972; Dworkin *et al.,* 1978; O'Shea and Cohen, 1972; Weinstein and Getz, 1978), research annotations and reviews (American Dental Association, 1973a, 1975a), proceedings of major conferences (Bryant *et al.,* 1979; Ingersoll *et al.,* 1977; Weinstein, 1978), and individual reports, articles, and monographs (Crain *et al.,* 1969;

Lambert and Freeman, 1967; Newman and Anderson, 1972; Sherlock and Morris, 1972).

Today, although we are far from the ultimate goal of the social and behavioral sciences in dentistry—that of understanding, explaining, and predicting oral health behavior—the literature suggests that there are several important social and behavioral science research contributions to oral health.

(1) Application of social survey technology, namely, the public opinion poll, questionnaire construction, and analysis:

Best known of the contributions from the social sciences is survey technology, which helps to define the parameters of the population being studied. Obtaining valid, dependable information from providers of dental care, health professions students or aspirants, consumers, nonconsumers, and administrators of care programs regarding backgrounds, attitudes, values, beliefs, and practices has become a commonplace first step in the research process. Sophisticated social survey techniques applied by the ADA's Bureau of Behavioral and Economic Research studies have shown an increase in the survey response rates of dentists (American Dental Association, 1973b, 1975b).

Survey research has also found that neither an increased level of knowledge about dental disease nor the removal of the income inhibitor will increase dental visit utilization to the anticipated goal (Douglas and Cole, 1979; Haefner, 1974). The associated analytic techniques of survey research have shown that many interrelated factors may act as motivators and inhibitors; some factors appear more closely related than others to the dependent variable of visit behavior, such as a person's perceived need for care (Cohen, 1978b). In addition to these illustrations, the social survey method also has been applied to studies of the dental student, career choice, fluoridation, and health education (Richards and Cohen, 1971). In efforts to assess the degree of public awareness, knowledge, and sentiment toward community water fluoridation, national public opinion data collected by the USPHS and the National Opinion Research Center have been of value to public health planners mapping national strategies for promotion (O'Shea and Cohen, 1966).

(2) Development of health education research, primarily that concerning influences of mothers on children and pressures of peer groups for conformity:

Perhaps the primary result of sociodental research has been to stress behavior rather than knowledge as the target for program planners. Rayner (1970) found, for example, that the primary

influence for change in a child's dental behavior was the mother's example—not what she said, but what she actually did. It follows that the next logical step toward influencing the child would be to learn how to influence the mother to explore her own motivations for actions. The work of Kegeles (1968) confirms certain propositions regarding the potential to change a person's psychosocial view of desired behavior through awareness of susceptibility, severity, alternative actions, and relative advantages or reduced threats of appropriate actions. The importance of this literature and more recent experimental studies is that there is evidence of patterns in dental behaviors and these patterns can be studied, predicted, and changed.

(3) Orientation toward problem analysis and problem solving in human relations:

The existence of disciplines devoted to the study of human relations has had an impact on dentistry, on its delivery, and on the way the subject is taught to students of the health professions. In the 1960s groups interested in sensitivity training and in participative management became active in dental schools. To a certain extent there was more evangelism than research associated with these early endeavors, but they paved the way for more recent work which related motivation to performance of students faculty, or patients (Kreit, 1971).

(4) Management by objectives and education by objectives, particularly in relation to dental curricula and the use of dental teams for treating patients:

Certain concepts, derived largely from the work of psychologist B. F. Skinner, have been of significance to dental education both for health professions training and for continuing education. These concepts include the behavioral definition of training goals, task analysis, and the systematic application of principles of reinforcement to the learning situation (Mackenzie, 1973; Mager, 1962).

(5) Testing procedures for recruitment of personnel and criteria for selecting dental and auxiliary students:

There is a long-standing contribution of the social and behavioral sciences to improved selection, placement, and assessment of personnel. In industry, perhaps it has been the most conspicuous and persistent manifestation of the scientific approach to human problems. Tests have been developed to provide clues to applicants' personality traits, intelligence levels, vocational aptitudes, and academic achievements (Lennon and Hollenbeck, 1972). The dental field has benefited by continually refining at-

tempts to apply scientific methodology to career selection and training (Mackenzie, 1971). While testing methods at this stage in their development may not be totally free of cultural bias, they are designed with a view toward building more objective and sound procedures.

The schema related to linking communities needing dentists with dentists needing practice locations has also proved beneficial (Born and Burton, 1978). Although this idea was originally planned for one state and then extended to the region, the ADA and the Kellogg Foundation have supported its further development for national application.

(6) Application of situation analyses to facilitate planning for large public efforts, such as community water fluoridation and national delivery of health care:

In studying a model for the delivery of either comprehensive dental services or for individual services, such as community-based fluoride programs, it is necessary to define, quantitatively and qualitatively, all the actors and elements involved in the community. Representatives of providers, consumers, and administrators constitute at least three major actor groups located at the core of such a potential model. The oral health goods and services exchanged among these groups constitute the elements, as are the channels of communication used to exchange those goods and services. No matter how one tries to define the inclusive nature of the dental community, the concept defies geographic boundaries, such as state and county lines. Model building in the social sciences has provided a conceptual starting point for the International Collaborative Study of Dental Manpower Systems in Relation to Oral Health Status, now operatively analyzing the dental-care delivery status of approximately a dozen countries (Cohen, 1978b). As such models are refined, modified, and improved, the impact of social and behavioral analysis in dentistry becomes more evident to planners in the dental community.

These contributions to dentistry from social and behavioral research are abbreviated and represent only the first stages of an entry into a world of continually increasing demands. Our previously stated goal to increase the level of understanding sufficiently to explain and predict human behavior in relation to oral health requires that we cross new frontiers to formulate policy and action alternatives rooted in a base of scientifically derived information.

Essentially, we have not spent much time developing new theories or testing previously developed theories. Most of our work has been empirical,

and although the information is scientifically gathered and analyzed, it is generally nontheoretical. Our technologies for promoting health should be tested against available theoretically based hypotheses drawn from communication systems, social systems, and socialization theories. Presently we have a series of discrete research products that are hard to relate to one another.

A social-problems approach to this research is needed, since the goal of this conference is devoted to improving oral health behavior by reaching the individual through other individuals, group structures, and organization and community mechanisms. The idea of a social-problems approach is neither original nor invented by only one person, as it has been used previously by a few members of the sociodental community.

A social problem is defined as any condition that causes strain, tension, conflict, or frustration and interferes with the fulfilment of a need. The dimensions of a social problem are psychological because there is emotional or intellectual disturbance at the individual level and are *sociological* because relief or a solution comes from social changes in the situation or a modification of the environment. The value of considering oral disease from this perspective is that, theoretically, one could hope to discover possible changes that would not invade individual rights, but could alter structures in the environment to prevent the proliferation of the problem itself.

The interrelations between human health behavior and the culture of society is systematically studied in the light of a variety of social and behavioral science disciplines. Although each discipline may approach the problem by employing somewhat different methods and points of view, their goals are similar: to define the selected behavior and to determine the processes by which it is induced, maintained, or altered by a host of sociocultural variables. The underlying assumptions are that when the human behavior fails to match the surrounding environmental structures, the discrepancy is "undesirable." Increasingly, those who use the social-problems approach have seen how problems may be attributed less to the individual than to the surrounding forces. The idea is to move away from the study of the person caught up in the problem—the alcoholic, the poverty stricken—and study the conditions causing the problem in search of a solution that might lead to a remedy.

A deliberate emphasis on social and environmental structures can offer possibilities for a larger, more efficient scale of social change, and thus it might be more appropriate to apply a social-problems approach rather than an individual-problem approach. We would no longer assume that the interpersonal nexus is the only possible mechanism intervening between one's position in the socioenvironmental context and one's personal health behavior. We should study the effects of the socioenviron-

ment, regardless of whether or not these factors are mediated by interpersonal relationships. Consider routine personal hygiene, for example, as it relates to the demands based on job responsibilities, time pressures, and the ability to take time off from work. Structural factors can prevent the practice of optimal personal hygiene; these are considered "structural" factors because they are built into the job and do not arise because individuals do not wish to take the time to be "good dental citizens." In theory, the answer to problems of individual behavior may be found in various environmental factors, providing potential opportunities for collaboration of several disciplines.

These assumptions are not presented to decry the need for research on individual behavior, as sociodental research still has such a need. They are offered, instead, in the hope that a public health approach will be adopted by those who synthesize the disparate and discrete contributions of the past and in the hope that the ultimate dependent variable of their research will be the oral health status of groups of people. Although many "one-to-one" intervention strategies can be effective, their costliness becomes prohibitive in the context of delivering health care to large groups of people. Instead, intervention strategies that depend upon institutional, group, or community involvement could be more cost effective. We should, therefore, promote social research and development focused on multiinstitutional or sociostructural factors for improving the status of oral health.

One way to explore a new frontier, free from our own disciplinary yokes, is to assume that the answer to "social problems" may not lie totally within our own culture. We need to test hypotheses and theories in varying social contexts and to create new hypotheses on the basis of a broader range of different experiences related to dentistry. This could be achieved within this country by varying the social context while keeping the intervention strategy constant or by exploring new frontiers by engaging in cross-national research. Although defining the social context and relating it to oral health behaviors has never been addressed adequately in the sociodental literature, such research conceivably may provide useful clues for problem resolution.

As investigators adopt the social-problems approach, the need for the broadest possible range of disciplinary skills becomes apparent. We do not yet know all the reasons why people adopt personal preventive practices or adopt community health measures. It is necessary to look beyond interpersonal relationships, as these answers may lie in community institutions, in family contexts, in legislative domains, and in administrative structures.

Finally, we must make better use of collaborative scientific teams, despite the difficulties involved in arranging and sustaining them. Incor-

porating clinical questions asked by either providers or consumers may be a beneficial step in forming our hypotheses. Previously, each researcher in his own discipline has posed questions relevant to his own field, but a combined effort by social *and* dental researchers might prove more productive if they would join together to raise questions.

The research advanced by working together across disciplines and cultures might result in a future when no critic will have occasion to ask us "science. . . for what?" The challenge of this conference is to pose appropriate specific research questions which will enable individual behavior and social change to occur for the purpose of achieving optimum oral health.